

EXP 3 Construct Min and Max Heap using arrays, delete any element and display the content of the Heap.

AIM:

To create Min Heap and Max Heap using arrays, delete an element, and display the heaps.

PROCEDURE:

1. Enter the number of elements.
2. Read the elements into an array.
3. Create the Min Heap by inserting each element and maintaining the min-heap property.
4. Create the Max Heap by inserting each element and maintaining the max-heap property.
5. Display both the Min and Max heap.
6. Enter the element to delete.
7. Delete the element from both heaps.
8. Display the updated Min Heap and Max Heap.

PROGRAM:

```
#include <stdio.h>
void swap(int *a, int *b)
{
    int t = *a;
    *a = *b;
    *b = t;
}

void minHeap(int a[], int n)
{
    int i, p;
    for (i = 1; i < n; i++)
    {
        int c = i;
        while (c > 0)
        {
            p = (c - 1) / 2;
            if (a[p] > a[c])
            {
                swap(&a[p], &a[c]);
                c = p;
            }
            else
                break;
        }
    }
}
```

```

void maxHeap(int a[], int n)
{
    int i, p;
    for (i = 1; i < n; i++)
    {
        int c = i;
        while (c > 0)
        {
            p = (c - 1) / 2;
            if (a[p] < a[c])
            {
                swap(&a[p], &a[c]);
                c = p;
            }
            else
                break;
        }
    }
}

```

```

void deleteElement(int a[], int *n, int key, int isMin)
{
    int i;

    // Find the element
    for (i = 0; i < *n; i++)
    {
        if (a[i] == key)
            break;
    }

    if (i == *n)
    {
        printf("Element not found\n");
        return;
    }

    // Replace with last element
    a[i] = a[*n - 1];
    (*n)--;

    // Rebuild heap
    if (isMin)
        minHeap(a, *n);
    else
        maxHeap(a, *n);
}

```

```

}
//DISPLAY
void display(int a[], int n)
{
    int i;
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);
    printf("\n");
}

int main()
{
    int n, i;
    int min[20], max[20];

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements: ");
    for (i = 0; i < n; i++)
    {
        scanf("%d", &min[i]);
        max[i] = min[i];
    }

    int minSize = n;
    int maxSize = n;

    minHeap(min, minSize);
    maxHeap(max, maxSize);

    printf("\nMin Heap: ");
    display(min, minSize);

    printf("Max Heap: ");
    display(max, maxSize);

    int key;
    printf("\nEnter element to delete: ");
    scanf("%d", &key);

    deleteElement(min, &minSize, key, 1);
    deleteElement(max, &maxSize, key, 0);

    printf("\nMin Heap after deletion: ");
    display(min, minSize);

```

```
    printf("Max Heap after deletion: ");  
    display(max, maxSize);  
  
    return 0;  
}
```

OUTPUT:

D:\>gcc Heap.c -o Heap.exe

D:\>Heap.exe

Enter number of elements: 6

Enter elements: 10 20 5 6 1 8

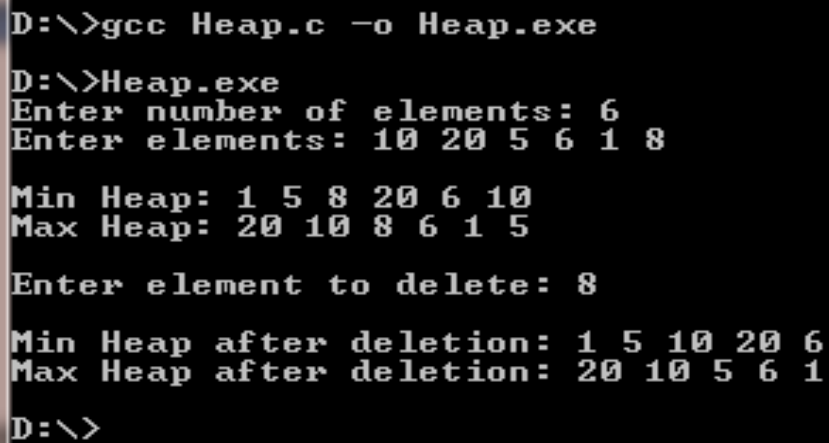
Min Heap: 1 5 8 20 6 10

Max Heap: 20 10 8 6 1 5

Enter element to delete: 8

Min Heap after deletion: 1 5 10 20 6

Max Heap after deletion: 20 10 5 6 1



```
D:\>gcc Heap.c -o Heap.exe  
D:\>Heap.exe  
Enter number of elements: 6  
Enter elements: 10 20 5 6 1 8  
  
Min Heap: 1 5 8 20 6 10  
Max Heap: 20 10 8 6 1 5  
  
Enter element to delete: 8  
  
Min Heap after deletion: 1 5 10 20 6  
Max Heap after deletion: 20 10 5 6 1  
  
D:\>
```